

HW9

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Exercise 9.1

设随机变量 $X \sim U(a, b)$, 对 $k = 1, 2, 3, 4$, 求 $\mu_k = E(X^k)$ 与 $\nu_k = E(X - E(X))^k$. 进一步求此分布的偏度系数和峰度系数.

Proof. 对于任意的 $k \in \mathbb{Z}_+$,

$$\begin{aligned}\mu_k &= E(X^k) = \int X^k dP \\ &= \int x^k dP_X \\ &= \int_a^b x^k \frac{1}{b-a} dx \\ &= \frac{1}{k+1} (b^{k+1} - a^{k+1}) \frac{1}{b-a} \\ &= \frac{1}{k+1} \frac{b^{k+1} - a^{k+1}}{(b-a)}\end{aligned}$$

$$\nu_k = (E(X - E(X))^k)$$

$$E(X) = \frac{1}{2}(b+a), \quad Y = X - E(X) \sim U\left(\frac{1}{2}(a-b), \frac{1}{2}(b-a)\right)$$

于是

$$\nu_k = \frac{1}{k+1} \frac{1}{2^{k+1}} \frac{(b-a)^{k+1} + (-1)^k (b-a)^{k+1}}{b-a} = \frac{1 + (-1)^k}{2^{k+1}(k+1)} (b-a)^k$$

带入 $k = 1, 2, 3, 4$ 即可.

偏度系数为

$$\beta_s = \frac{\nu_3}{(\nu_2)^{\frac{3}{2}}} = 0$$

峰度系数为

$$\beta_k = \frac{\nu_4}{(\nu_2)^2} = \frac{\frac{1}{245}(b-a)^4}{\left(\frac{1}{12}(b-a)^2\right)^2} = \frac{2^4 3^2}{2^4 5} = \frac{9}{5}$$

□

Exercise 9.2

求以下分布的中位数:

1. 区间 (a, b) 上的均匀分布;
2. 正态分布 $N(\mu, \sigma^2)$;
3. 对数正态分布 $LN(\mu, \sigma^2)$.

Proof. 1. 由

$$F(x_{0.5}) = \int_a^{x_{0.5}} \frac{1}{b-a} dx = 0.5$$

可得

$$\frac{x_{0.5} - a}{b - a} = \frac{1}{2} \implies x_{0.5} = \frac{1}{2}(a + b)$$

2. 由正态分布的对称性,

$$P(X \leq \mu) = P(X \geq \mu) = 0.5$$

因此

$$x_{0.5} = \mu$$

3. 设 $Z \in LN(\mu, \sigma^2)$, 则 $Y = \log Z \sim N(\mu, \sigma^2)$

$$P(Z \leq x_{0.5}) = 0.5 \implies P(Y \leq \log x_{0.5}) = 0.5 \implies \log x_{0.5} = \mu$$

因此

$$x_{0.5} = e^\mu$$

□

Exercise 9.3

设随机变量 $X \sim \text{Exp}(\lambda)$, 对 $k = 1, 2, 3, 4$, 求 $\mu_k = E(X^k)$ 与 $\nu_k = E(X - E(X))^k$. 进一步求此分布的变异系数, 偏度系数和峰度系数.

Solution.

$$\mu_k = \int_0^{\infty} x^k \lambda e^{-\lambda x} dx = \frac{1}{\lambda^k} \int_0^{\infty} (\lambda x)^k e^{-\lambda x} d\lambda x = \frac{\Gamma(k+1)}{\lambda^k}$$

于是

$$\mu_1 = \frac{1}{\lambda}, \quad \mu_2 = \frac{2}{\lambda^2}, \quad \mu_3 = \frac{6}{\lambda^3}, \quad \mu_4 = \frac{24}{\lambda^4}$$

$$\nu_1 = 0, \quad \nu_2 = E(X^2) - (E(X))^2 = \mu_2 - \mu_1^2 = \frac{1}{\lambda^2}$$

$$\begin{aligned} \nu_3 &= E((X - \mu_1)^3) \\ &= E(X^3 - 3\mu_1 X^2 + 3\mu_1^2 X - \mu_1^3) \\ &= \mu_3 - 3\mu_1 \mu_2 + 3\mu_1^3 - \mu_1^3 \\ &= \frac{6}{\lambda^3} - \frac{6}{\lambda^3} + \frac{2}{\lambda^3} \\ &= \frac{2}{\lambda^3} \end{aligned}$$

$$\begin{aligned} \nu_4 &= E[(X - \mu_1)^4] \\ &= E(X^4 - 4\mu_1 X^3 + 6\mu_1^2 X^2 - 4\mu_1^3 X + \mu_1^4) \\ &= \mu_4 - 4\mu_1 \mu_3 + 6\mu_1^2 \mu_2 - 4\mu_1^4 + \mu_1^4 \\ &= \frac{24}{\lambda^4} - \frac{24}{\lambda^4} + \frac{12}{\lambda^4} - \frac{3}{\lambda^4} \\ &= \frac{9}{\lambda^4} \end{aligned}$$

变异系数为

$$C_v(X) = \frac{\sqrt{\nu_2}}{\mu_1} = 1$$

偏度系数为

$$\beta_s = \frac{\nu_3}{(\nu_2)^{\frac{3}{2}}} = 2$$

峰度系数为

$$\beta_k = \frac{\nu_4}{(\nu_2)^2} = 9$$

◇

Exercise 9.4

设随机变量 X 服从正态分布 $N(10, 9)$, 试求 $x_{0.1}$ 和 $x_{0.9}$.

Solution. 令 $Y = \frac{X-10}{3}$, 则 $Y \sim N(0, 1)$.

$$0.1 = P(X \leq x_{0.1}) = P\left(Y \leq \frac{x_{0.1} - 10}{3}\right)$$

于是

$$\frac{x_{0.1} - 10}{3} = \Phi^{-1}(0.1) \implies x_{0.1} = 3\Phi^{-1}(0.1) + 10 \approx 6.1552$$

类似地,

$$x_{0.9} = 3\Phi^{-1}(0.9) + 10 \approx 13.8448$$

◇

Exercise 9.5

设随机变量 X 服从双参数韦布尔分布, 其分布函数为

$$F(x) = 1 - \exp\left\{-\left(\frac{x}{\eta}\right)^m\right\}, \quad x > 0,$$

其中 $\eta > 0, m > 0$. 试写出该分布的 p 分位数 x_p 的表达式, 且求出当 $m = 1.5, \eta = 1000$ 时的 $x_{0.1}, x_{0.5}, x_{0.8}$ 的值.

Solution. 根据分位数的定义

$$F(x_p) = p$$

具体地

$$1 - \exp\left(-\left(\frac{x_p}{\eta}\right)^m\right) = p$$

得到

$$\exp\left(-\left(\frac{x_p}{\eta}\right)^m\right) = 1 - p \implies x_p = \eta(-\log(1 - p))^{\frac{1}{m}}$$

当 $m = 1.5, \eta = 1000$ 时,

$$x_p = 1000(-\log(1 - p))^{\frac{2}{3}}$$

带入数值, 得到

$$x_{0.1} \approx 223.14 \quad x_{0.5} \approx 781.80, \quad x_{0.8} \approx 1373.74$$

◇

Exercise 9.6

设随机变量 X 的分布密度函数 $p(x)$ 关于直线 $x = c$ 是对称的, 且 $E(X)$ 存在, 试证:

1. 这个对称点 c 既是均值又是中位数, 即 $E(X) = x_{0.5} = c$;
2. 如果 $c = 0$, 则 $x_\alpha = -x_{1-\alpha}$.

Proof. 1. 我们有

$$p(x) = p(2c - x)$$

于是

$$\int_{-\infty}^c p(x) dx = \int_{-\infty}^c p(2c - x) dx = \int_c^{\infty} p(x) dx$$

又

$$1 = \int_{-\infty}^{\infty} p(x) dx = \int_{-\infty}^c p(x) dx + \int_c^{\infty} p(x) dx$$

故

$$F(c) = 0.5, \quad c = x_{0.5}$$

此外,

$$\begin{aligned}
 E(X - c) &= \int_{-\infty}^{\infty} (x - c) p(x) dx \\
 &= \int_{-\infty}^{\infty} (x - c) p(2c - x) dx \\
 &= \int_{-\infty}^{\infty} (c - x) p(x) dx \\
 &= - \int_{-\infty}^{\infty} (x - c) p(x) dx = -E(X - c)
 \end{aligned}$$

故 $E(X - c) = 0$, $E(X) = c$.

2. 若 $c = 0$, 则 $p(x) = p(-x)$,

$$\begin{aligned}
 \alpha = F(x_\alpha) &= \int_{-\infty}^{x_\alpha} p(x) dx \\
 &= \int_{-\infty}^{x_\alpha} p(-x) dx \\
 &= \int_{-x_\alpha}^{\infty} p(x) dx \\
 &= 1 - F(-x_\alpha)
 \end{aligned}$$

于是

$$F(-x_\alpha) = 1 - \alpha \implies -x_\alpha = x_{1-\alpha}$$

即 $x_\alpha = -x_{1-\alpha}$

□

Exercise 9.7

试证随机变量 X 的偏度系数与峰度系数对位移和改变比例尺是不变的, 即对任意的实数 $a, b(b \neq 0)$, $Y = a + bX$ 与 X 有相同的偏度系数与峰度系数.

Proof. 设 $\mu_k = E(X^k)$, $\nu_k = E[(X - E(X))^k]$, $\mu'_k = E(Y^k)$, $\nu'_k = E[(Y - E(Y))^k]$. 则

$$\mu'_1 = E(a + bX) = a + b\mu_1, \quad \nu'_1 = E((Y - \mu_1)^2) = b^2\nu_1$$

$$\nu'_k = E[(Y - E(Y))^k] = E[((a + bX) - (a + b\mu_1))^k] = E[b^k(X - \mu_1)^k] = b^k \nu_k$$

设 β_s, β_k 为 X 的偏度系数和峰度系数, β'_s, β'_k 为 Y 的偏度系数和峰度系数. 则

$$\beta'_s = \frac{\nu'_3}{(\nu'_2)^{\frac{3}{2}}} = \frac{b^3 \nu_3}{(b^2 \nu_1)^{\frac{3}{2}}} = \frac{\nu_3}{\nu_1} = \beta_s$$

$$\beta'_k = \frac{\nu'_4}{(\nu'_2)^2} = \frac{b^4 \nu_4}{(b^2 \nu_2)^2} = \frac{\nu_4}{(\nu_2)^2} = \beta_k$$

□

Exercise 9.8

某种绝缘材料的使用寿命 T (单位: 小时) 服从对数正态分布 $LN(\mu, \sigma^2)$. 若已知分位数 $t_{0.2} = 5000$ 小时, $t_{0.8} = 65000$ 小时, 求 μ 和 σ .

Solution. 设 $X = \log T$, 则 $X \sim N(\mu, \sigma^2)$ 令 $Y = \frac{X - \mu}{\sigma}$, 则 $Y \sim N(0, 1)$. 我们有

$$0.2 = P(T \leq 5000) = P(X \leq \log 5000) = P\left(Y \leq \frac{\log 5000 - \mu}{\sigma}\right)$$

于是

$$\Phi^{-1}(0.2) = \frac{\log 5000 - \mu}{\sigma}$$

类似地,

$$\Phi^{-1}(0.8) = \frac{\log 65000 - \mu}{\sigma}$$

两式相减, 得到

$$\Phi^{-1}(0.8) - \Phi^{-1}(0.2) = \frac{\log 13}{\sigma}$$

于是

$$\sigma = \frac{\log 13}{\Phi^{-1}(0.8) - \Phi^{-1}(0.2)} \approx 1.52381$$

进而

$$\mu = \log 5000 - \Phi^{-1}(0.2) \sigma = \log 5000 - \Phi^{-1}(0.2) \frac{\log 13}{\Phi^{-1}(0.8) - \Phi^{-1}(0.2)} \approx 9.79967$$

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Exercise 9.9

某厂决定按过去生产状况对月生产额最高的 5% 的工人发放高产奖. 已知过去每人每月生产额 X (单位: 千克) 服从正态分布 $N(4000, 60^2)$, 试问高产奖发放标准应把生产额定为多少?

Solution. 应该把生产额定为 $x_{0.95}$. 令 $Y = \frac{X-4000}{60}$, 则 $Y \sim N(0, 1)$.

$$0.95 = P(X \leq x_{0.95}) = P\left(Y \leq \frac{x_{0.95} - 4000}{60}\right)$$

于是

$$\frac{x_{0.95} - 4000}{60} = \Phi^{-1}(0.95)$$

从而生产额定为

$$x_{0.95} = 60\Phi^{-1}(0.95) + 4000 \approx 4000 + 60 \times 1.645 = 4098.7$$

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Exercise 9.10

100 件产品中有 50 件一等品, 30 件二等品, 20 件三等品. 从中任取 5 件, 以 X, Y 分别表示取出的 5 件中一等品、二等品的件数, 在以下情况下求 (X, Y) 的联合分布列:

1. 不放回抽取;
2. 有放回抽取.

Solution. 1. 当进行不放回抽样时, 对于 $k, l \in \mathbb{Z}_{\geq 0}$ 使得 $k + l \leq 5$, 我们有

$$P((X, Y) = (k, l)) = \frac{C_{50}^k \cdot C_{30}^l \cdot C_{20}^{5-k-l}}{C_{100}^5}$$

k \ l		0	1	2	3	4	5
计算得到	0	0.00021	0.00193	0.00659	0.01025	0.00728	0.00189
	1	0.00322	0.02271	0.05489	0.05393	0.01820	0.00000
	2	0.01855	0.09274	0.14155	0.06606	0.00000	0.00000
	3	0.04946	0.15620	0.11325	0.00000	0.00000	0.00000
	4	0.06118	0.09177	0.00000	0.00000	0.00000	0.00000
	5	0.02814	0.00000	0.00000	0.00000	0.00000	0.00000

2. 当进行放回抽样时, 对于 $k, l \in \mathbb{Z}_{\geq 0}$ 使得 $k + l \leq 5$, 我们有

$$\begin{aligned} P((X, Y) = (k, l)) &= C_5^{k+l} C_{k+l}^k \frac{1}{2^k} \left(\frac{3}{10}\right)^l \left(\frac{1}{5}\right)^{5-k-l} \\ &= \frac{5!}{k!l!(5-k-l)!} \frac{1}{2^k} \left(\frac{3}{10}\right)^l \left(\frac{1}{5}\right)^{5-k-l} \end{aligned}$$

$k \setminus l$	0	1	2	3	4	5
0	0.00032	0.00240	0.00720	0.01080	0.00810	0.00243
1	0.00400	0.02400	0.05760	0.05400	0.02025	0.00000
计算得到 2	0.02000	0.09000	0.13500	0.08100	0.00000	0.00000
3	0.05000	0.15000	0.11250	0.00000	0.00000	0.00000
4	0.06250	0.09375	0.00000	0.00000	0.00000	0.00000
5	0.03125	0.00000	0.00000	0.00000	0.00000	0.00000

◇

Exercise 9.11

盒子里装有 3 个黑球, 2 个红球, 2 个白球, 从中任取 4 个, 以 X 表示取到黑球的个数, 以 Y 表示取到红球的个数, 试求 $P(X = Y)$.

Solution.

$$P((X, Y) = (k, k)) = \frac{C_3^k C_2^k C_2^{4-2k}}{C_7^4}, \quad k = 1, 2$$

当 k 为其他整数时, $P((X, Y) = (k, k)) = 0$. 于是

$$\begin{aligned} P(X = Y) &= P((X, Y) = (1, 1)) + P((X, Y) = (2, 2)) \\ &= \frac{6}{35} + \frac{3}{35} \\ &= \frac{9}{35} \end{aligned}$$

◇

Exercise 9.12

设随机变量 $X_i, i = 1, 2$, 的分布列如下, 且满足 $P(X_1 X_2 = 0) = 1$, 试求 $P(X_1 = X_2)$.

X_i	-1	0	1
P	0.25	0.5	0.25

Proof. 由于 $P(X_1 X_2 = 0) = 1$, 我们有

$$P(X_1 X_2 = 0) = P(X_1 = 0) + P(X_2 = 0) - P(X_1 = X_2 = 0)$$

因此

$$P(X_1 = X_2 = 0) = 1 - 0.5 - 0.5 = 0$$

从而

$$P(X_1 = X_2) = P(X_1 = X_2 = -1) + P(X_1 = X_2 = 1)$$

但是

$$\{X_1 = X_2 = -1\} \subseteq \{X_1 X_2 \neq 0\}, \quad \{X_1 = X_2 = 1\} \subseteq \{X_1 X_2 \neq 0\}$$

因此

$$P(X_1 = X_2) \leq 2P(X_1 X_2 \neq 0) = 2(1 - P(X_1 X_2 = 0)) = 0$$

故 $P(X_1 = X_2) = 0$. □

Exercise 9.13

设随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} k(6 - x - y), & 0 < x < 2, \quad 2 < y < 4, \\ 0, & \text{其他.} \end{cases}$$

试求:

1. 常数 k ;
2. $P(X < 1, Y < 3)$;
3. $P(X < 1.5)$;
4. $P(X + Y \leq 4)$.

Proof. 1. 我们有

$$\int_{\mathbb{R}^2} p(x, y) \, dx \, dy = 1$$

从而

$$1 = k \int_0^2 \int_2^4 (6 - x - y) \, dy \, dx = k \int_0^2 (6 - 2x) \, dx = 8k$$

$$\text{故 } k = \frac{1}{8}$$

2.

$$P(X < 1, Y < 3) = \int_0^1 \int_2^3 \frac{1}{8} (6 - x - y) dx dy = \frac{1}{8} \int_0^1 \frac{7}{2} - x dx = \frac{3}{8}$$

3.

$$P(X < 1.5) = \int_0^{1.5} \int_2^4 \frac{1}{8} (6 - x - y) dx dy = \frac{1}{8} \int_0^{1.5} (6 - 2x) dx = \frac{27}{32}$$

4.

$$\begin{aligned} P(X + Y \leq 4) &= \int_0^2 \int_2^{4-x} \frac{1}{8} (6 - x - y) dy dx \\ &= \frac{1}{8} \int_0^2 (6 - x)(2 - x) - \frac{1}{2} ((4 - x)^2 - 4) dx \\ &= \frac{1}{8} \int_0^2 \left(6 - 4x + \frac{x^2}{2} \right) dx \\ &= \frac{2}{3} \end{aligned}$$

□

Exercise 9.14

设二维随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} 4xy, & 0 < x < 1, \quad 0 < y < 1, \\ 0, & \text{其他.} \end{cases}$$

试求:

1. $P(0 < X < 0.5, 0.25 < Y < 1)$;
2. $P(X = Y)$;
3. $P(X < Y)$;
4. (X, Y) 的联合分布函数.

Solution. 1.

$$\begin{aligned}
 P(0 < X < 0.5, 0.25 < Y < 1) \\
 &= \int_0^{0.5} \int_{0.25}^1 4xy \, dy \, dx \\
 &= \int_0^{\frac{1}{2}} \frac{15}{8} x \, dx \\
 &= \frac{15}{64}
 \end{aligned}$$

2.

$$\begin{aligned}
 P(X = Y) &= \int \chi_{\{X=Y\}} \, dP \\
 &= \int_{(0,1)^2} \chi_{\{x=y\}} (\, dx \times dy) \\
 &= \lambda(\{(x, y) \in (0, 1)^2 : x = y\})
 \end{aligned}$$

其中 λ 表示 2 维 Lebesgue 测度. 由于集合 $\{(x, y) \in (0, 1)^2 : x = y\}$ 是一个 2-Lebesgue 零测集, 我们有

$$P(X = Y) = 0$$

3.

$$P(X < Y) = \int_0^1 \int_0^y 4xy \, dx \, dy = \int_0^1 2y^3 \, dy = \frac{1}{2}$$

4. 若 $s \leq 0$ 或 $t \leq 0$, 则 $F_{(X,Y)}(s, t) = 0$, 若 $s, t \geq 1$, 则 $F_{(X,Y)} = 1$. 若 $0 < s < 1, t \geq 1$, 则

$$\begin{aligned}
 F_{(X,Y)}(s, t) &= \int_0^s \int_0^1 4xy \, dy \, dx \\
 &= \int_0^s 2x \, dx \\
 &= s^2
 \end{aligned}$$

若 $0 < t < 1, s \geq 1$, 则

$$F_{(X,Y)}(s, t) = \int_0^1 \int_0^t 4xy \, dy \, dx = t^2$$

若 $0 < t, s < 1$, 则

$$\begin{aligned} F_{(X,Y)}(s, t) &= \int_0^s \int_0^t 4xy \, dy \, dx \\ &= \int_0^s 2xt^2 \, dx \\ &= s^2 t^2 \end{aligned}$$

综上所述

$$F_{(X,Y)}(s, t) = \begin{cases} 0, & s \leq 0 \text{ or } t \leq 0 \\ s^2 t^2, & 0 < s, t < 1 \\ s^2, & 0 < s < 1, t \geq 1 \\ t^2, & 0 < t < 1, s \geq 1 \\ 1, & s, t \geq 1 \end{cases}$$

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Exercise 9.15

设二维随机变量 (X, Y) 在边长为 2, 中心为 $(0, 0)$ 的正方形区域内服从均匀分布, 试求 $P(X^2 + Y^2 \leq 1)$.

Solution. 我们有

$$p_{(X,Y)} = \begin{cases} \frac{1}{4}, & -1 < x < 1, \quad -1 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned}
P(X^2 + Y^2 \leq 1) &= \int \chi_{\{X^2 + Y^2 \leq 1\}} dP \\
&= \int \chi_{\{x^2 + y^2 \leq 1\}} p_{(X,Y)} d\lambda \\
&= \frac{1}{4} \int_{\mathbb{R}^2} \chi_{\{x^2 + y^2 \leq 1\}} d\lambda \\
&= \frac{\pi}{4}
\end{aligned}$$

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Exercise 9.16

设二维随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} k, & 0 < x^2 < y < x < 1, \\ 0, & \text{其他.} \end{cases}$$

1. 试求常数 k ;
2. 求 $P(X > 0.5)$ 和 $P(Y < 0.5)$.

Solution. 1.

$$\begin{aligned}
1 &= \int_{\mathbb{R}^2} k \chi_{\{0 < x^2 < y < x < 1\}} d\lambda \\
&= \int_0^1 \int_0^1 k \chi_{\{0 < x^2 < y < x < 1\}} dy dx \\
&= \int_0^1 \int_{x^2}^x k dy dx \\
&= \int_0^1 k(x - x^2) dx \\
&= k \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{k}{6}
\end{aligned}$$

故 $k = 6$

2.

$$P(X > 0.5) = \int_{0.5}^1 \int_{x^2}^x 6 dy dx = 6 \int_{0.5}^1 (x - x^2) dx = \frac{1}{2}$$

$$\begin{aligned}
P(Y < 0.5) &= \int_0^1 \int_0^{0.5} 6\chi_{\{0 < x^2 < y < x < 1\}} dy dx \\
&= \int_0^{0.5} \int_0^1 6\chi_{\{0 < y < x < \sqrt{y} < 1\}} dx dy \\
&= \int_0^{0.5} \int_y^{\sqrt{y}} 6 dx dy \\
&= \int_0^{\frac{1}{2}} 6(\sqrt{y} - y) dy \\
&= 6 \left[\frac{2}{3} y^{\frac{3}{2}} - \frac{1}{2} y^2 \right]_0^{\frac{1}{2}} \\
&= 6 \left(\frac{2}{3} \frac{1}{2\sqrt{2}} - \frac{1}{8} \right) \\
&= \sqrt{2} - \frac{3}{4}
\end{aligned}$$

◇

Exercise 9.17

设随机变量 Y 服从参数为 $\lambda = 1$ 的指数分布, 定义随机变量 X_k 如下:

$$X_k = \begin{cases} 0, & Y \leq k, \\ 1, & Y > k, \end{cases} \quad k = 1, 2.$$

求 X_1 和 X_2 的联合分布列.

Solution.

$$P((X_1, X_2) = (0, 0)) = P(Y \leq 1, Y \leq 2) = P(Y \leq 1) = \int_0^1 \lambda e^{-\lambda x} dx = 1 - e^{-1}$$

$$P((X_1, X_2) = (0, 1)) = P(Y \leq 0, Y > 2) = 0$$

$$P((X_1, X_2) = (1, 0)) = P(Y > 1, Y \leq 2) = \int_1^2 \lambda e^{-\lambda x} dx = e^{-1} - e^{-2}$$

$$\begin{aligned}
 P((X_1, X_2) = (1, 1)) &= P(Y > 1, Y > 2) = P(Y > 2) \\
 &= 1 - P(Y \leq 2) \\
 &= 1 - P(Y \leq 1) - P(1 \leq Y \leq 2) \\
 &= e^{-2}
 \end{aligned}$$

$X_1 \backslash X_2$	$X_2 = 0$	$X_2 = 1$
$X_1 = 0$	$1 - e^{-1}$	0
$X_1 = 1$	$e^{-1} - e^{-2}$	e^{-2}

◇

Exercise 9.18

设二维随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} e^{-y}, & 0 < x < y, \\ 0, & \text{otherwise.} \end{cases}$$

试求 $P(X + Y \leq 1)$.

Solution.

$$\begin{aligned}
 P(X + Y \leq 1) &= \int_0^\infty \int_0^\infty \chi_{\{x+y \leq 1, x < y\}} e^{-y} dy dx \\
 &= \int_0^{\frac{1}{2}} \int_x^{1-x} e^{-y} dy dx \\
 &= \int_0^{\frac{1}{2}} e^{-x} - e^{x-1} dx \\
 &= 1 - e^{-\frac{1}{2}} - e^{-\frac{1}{2}} + e^{-1} \\
 &= 1 - 2e^{-\frac{1}{2}} + e^{-1}
 \end{aligned}$$

◇

Exercise 9.19

设二维随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} 1/2, & 0 < x < 1, \quad 0 < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

求 X 与 Y 中至少有一个小于 0.5 的概率.

Proof.

$$\begin{aligned} P(X, Y > 0.5) &= \int_{0.5}^1 \int_{0.5}^2 \frac{1}{2} dy dx \\ &= \frac{3}{8} \end{aligned}$$

故 X, Y 中至少有一个小于 0.5 的概率为

$$P(\{X \leq 0.5\} \cup \{Y \leq 0.5\}) = 1 - P(X, Y > 0.5) = \frac{5}{8}$$

□

Exercise 9.20

从 $(0, 1)$ 中随机地取两个数, 求其积不小于 $3/16$, 且其和不大于 1 的概率.

Proof. 在 $(0, 1)$ 上,

$$1 - x > \frac{3}{16x} \iff (4x - 1)(4x - 3) < 0 \iff x \in \left(\frac{1}{4}, \frac{3}{4}\right)$$

$$\begin{aligned} \int_{(0,1)^2} \chi_{\{xy \geq \frac{3}{16}, x+y \leq 1\}} d\lambda &= \int_{\frac{1}{4}}^{\frac{3}{4}} \int_{\frac{3}{16x}}^{1-x} 1 dy dx \\ &= \int_{\frac{1}{4}}^{\frac{3}{4}} 1 - x - \frac{3}{16x} dx \\ &= \frac{1}{4} - \frac{3}{16} \ln 3 \end{aligned}$$

□